

or your **CKD Digital Twin / Cost of Doing Nothing Simulator**, you need a cost formula that increases with disease progression and major events (dialysis, hospitalization, transplant).

Recommended Cost Formula

$$\text{Annual CKD Cost} = C_{stage} + C_{hospital} + C_{dialysis} + C_{cvd} + C_{diabetes}$$

Where:

- **Cstage** = baseline cost according to CKD stage
- **Chospital** = hospitalization costs
- **Cdialysis** = dialysis costs (Stage 5)
- **Ccvd** = cardiovascular complication costs
- **Cdiabetes** = diabetes management costs

- **CKD Stage Cost Table (Annual Cost per Patient)**

CKD Stage	Approx. Annual Cost (USD)
Stage 1	\$3,000
Stage 2	\$5,000
Stage 3a	\$12,000
Stage 3b	\$20,000
Stage 4	\$35,000
Stage 5 (Pre-Dialysis)	\$50,000
Stage 5 + Dialysis	\$90,000
Kidney Transplant (Year 1)	\$120,000

Obtained from: [10.1007/s12325-023-02608-9](https://doi.org/10.1007/s12325-023-02608-9)

Cost Multipliers for Risk Factors

If using NHANES data:

Diabetes

$$C_{diabetes} = 0.15 \times C_{stage}$$

Hypertension

$$C_{HTN} = 0.10 \times C_{stage}$$

Cardiovascular Disease

$$C_{CVD} = 0.30 \times C_{stage}$$

Patients with diabetes and cardiovascular disease incur substantially higher healthcare costs.

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<https://doi.org/10.1186/s12913-024-11258-8>

Population-Level Formula

For a government simulator:

$$\text{Total Healthcare Cost} = \sum_{i=1}^n (C_{stage} + C_{hospital} + C_{dialysis} + C_{CVD})$$

Where n is the number of CKD patients.